POLYTECHNIQUE MONTRÉAL
affiliée à l'Université de Montréal

A lightweight canonicalization engine in C

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Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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A lightweight canonicalization engine in C

présenté par **William ZHANG**
en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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DEDICATION

To all the past, present and future CVXPY contributors

ACKNOWLEDGEMENTS

Texte / Text.

RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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LIST OF SYMBOLS AND ACRONYMS

AD	Algorithmic Differentiation
AML	Algebraic Modeling Language
ASA	Affine-Solver-Affine
CSC	Compressed Sparse Column
CSR	Compressed Sparse Row
DAG	Directed Acyclic Graph
DCP	Disciplined Convex Programming
DPP	Disciplined Parameterized Programming
LP	Linear Program
MOI	MathOptInterface
MPS	Mathematical Programming System
NLP	Nonlinear Program
QP	Quadratic Program
SOCP	Second-Order Cone Program
SDP	Semidefinite Program

CHAPTER 1 INTRODUCTION

Domain specific modeling languages for convex optimization such as CVXPY [1] and YALMIP [2] allow users to formulate optimization problems in a natural mathematical syntax. Through a process called *canonicalization*, these high-level problem descriptions are converted to a *standard conic form* used by many general-purpose convex solvers

$$\begin{array}{ll} \text{minimize} & c^T x + d \\ \text{subject to} & b - Ax \in K \end{array} \tag{1.1}$$

where $x \in \mathbf{R}^n$ is the decision variables, and the *problem data* are the matrix $A \in \mathbf{R}^{m \times n}$, the vectors $b \in \mathbf{R}^m$ and $c \in \mathbf{R}^n$, the scalar $d \in \mathbf{R}$ and the non-empty closed convex cone $K \subseteq \mathbf{R}^m$. Almost all of optimization practice can be divided into two groups, *modeling* and *solving*. The modeler's role is to formulate problems based on real-world business or engineering applications. The solver's role is to find an optimal solution given the data of a problem instance. Modeling languages serve as the communication bridge between these two groups. When improvements in solvers are made, modeling languages must adapt to allow users to benefit from them. Conversely, it is through natural formulations that modelers can express their requirements and help motivate solver improvements.

While the time taken to solve an optimization problem is often dominated by the solver, the time taken to *canonicalize* (or *compile*) a problem can also be significant, especially for large problems. In the case of CVXPY, the *compilation time* largely lies in *matrix stuffing*, the final step of canonicalization where symbolic expression trees are converted to explicit numerical matrices. Other modeling languages take different approaches to form the problem data, we review these *coefficient extraction* approaches in §2.4.

Atoms. Modeling languages for convex optimization are built on top of a library of functions which can be used to describe problems. These functions are known as *atoms* in CVXPY since they are the fundamental building blocks of the modeling language [3]. We write ϕ for a general atom from this library. Atoms have known attributes such as curvature, monotonicity, and sign, for example, .

Expression trees. An expression is recursively defined as an atom evaluated at a *subexpression*. The subexpression can either be a variable, a parameter, or another expression. Similarly to atoms, expressions can also be classified by curvature, monotonicity, and sign

which are determined by simple *composition rules* [4]. An expression tree whose atoms are all affine is known as an *affine expression tree*.

Parameterized conic form. The data inputs in a problem instance can either be fixed or varying. In CVXPY, varying data are referred to as *parameters* and fixed data are referred to as *constants*. Parameters are useful for specifying a *family* of problems where the structure is fixed [5]. A *parameterized cone program* extends the conic form in (1.1) by letting the problem data depend on a parameter vector $\theta \in \mathbf{R}^p$:

$$\begin{array}{ll} \text{minimize} & c(\theta)^T x + d(\theta) \\ \text{subject to} & b(\theta) - A(\theta)x \in \mathcal{K}. \end{array} \tag{1.2}$$

When c, d, b and A are *affine with respect to* θ, we can represent the problem data parameter dependence as

$$\begin{bmatrix} c \\ d \end{bmatrix} = Q \cdot \tilde{\theta}, \qquad \text{and} \qquad [A \quad b] = \sum_{i=1}^{p+1} \mathcal{R}[:,:,i] \cdot \tilde{\theta}_i, \tag{1.3}$$

where $Q \in \mathbf{R}^{(n+1)\times(p+1)}$, $\mathcal{R} \in \mathbf{R}^{m\times(n+1)\times(p+1)}$ and $\tilde{\theta} = (\theta, 1) \in \mathbf{R}^{p+1}$. When $A(\theta)$ is sparse, we also require that its structure or *sparsity pattern* must be fixed throughout the different iterations. We refer to Q as the objective *parameter map*, $\mathcal{R}$ as the constraint parameter map, and $\tilde{\theta}$ as the extended parameter vector. The equation for the constraint map is known as a *tensor* product, a generalization of matrix-vector product in higher dimensions. This notation is required to output a constraint matrix from a three dimensional tensor.

Matrix stuffing. Although the original user specified problem formulation may contain nonlinearities, after conversion to standard form, the *canonicalized problem* will only contain affine expressions (with respect to both parameters and variables). Concretely, matrix stuffing refers to the task of forming the constraint parameter map, as the objective parameter

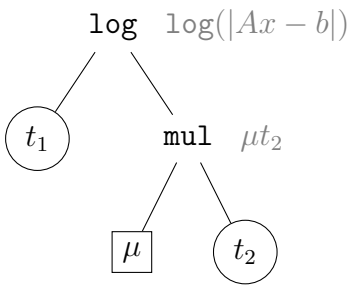


Figure 1.1 Example of expression tree for the function `log(|Ax-b|)`.

map is a scalar special case of it. Once the parameter maps are formed, the parameter dependence of the model is *cached* and doesn't rely on the expression tree representation anymore. Subsequent problem instances will only need to update θ and apply the maps to get the new problem data. We review current CVXPY matrix stuffing approaches and give examples of how the parameter maps are formed in §2.2.

Caching versus recompilation. Once a problem is parameterized, there are two broad strategies for handling repeated solves. The default approach in CVXPY, enabled by the DPP composition rules, forms the parameter maps once and *caches* them; subsequent solves merely apply the maps to a new θ. This is fast to re-solve but requires the problem to be DPP-compliant and stores the large three-dimensional tensor $\mathcal{R}$ in memory. The alternative, selected in CVXPY with `ignore_dpp=True`, skips the parameter maps entirely: parameters are treated as ordinary constants and the problem is re-canonicalized from scratch on every solve. This removes the DPP restriction and the tensor, but pays the full matrix-stuffing cost each time. Our method occupies the middle ground: like the `ignore_dpp` path it forms no parameter tensor and imposes no DPP restriction, but it caches the *sparsity pattern* of the problem data once and only refills the numerical values on each solve, recovering the amortized speed of the DPP path.

Contributions. We make the following contributions:

1. We propose an alternative method to matrix stuffing that doesn't rely on forming parameter maps. Instead, we treat matrix stuffing as a sparse Jacobian evaluation which naturally separates the symbolic structure of the Jacobian from its numerical values. This generalizes the `ignore_dpp` path—it applies to the same broad class of problems—while caching the sparsity pattern rather than re-deriving it on every solve.

2. We use the affine atoms from the `SparseDiffengine` package and its Python bindings to implement a canonicalization backend for CVXPY.

3. We benchmark our method against existing approaches and make improvements to CVXPY's codebase.

All code is open source and available at `https://github.com/cvxpy/cvxpy`.

Thesis outline. The rest of this thesis begins with a review of the literature on modeling languages, canonicalization, and matrix stuffing in §2. In §3 we describe a set of relevant

affine atoms and their implementation in a sparse differentiation package. In §4 we present benchmark results comparing our method to existing approaches. Finally, in §5 we summarize our contributions and discuss future improvements to the canonicalization backend.

CHAPTER 2 LITERATURE REVIEW

History and background Since the early days of linear programming (LP), practitioners have needed to translate mathematical formulations into data that solvers can process. At first, this meant encoding each coefficient of the problem matrices by hand, a process that is both tedious and error-prone [6]. *Matrix generators* were introduced in the late 1960s to automate this task: they compute the coefficients through explicit loops and write them out in solver-readable formats like MPS [7, 8].

Despite matrix generators removing most of the manual labor, a gap remained. Modelers naturally think in terms of sets, indices, and algebraic expressions, while procedural code constructs the coefficient matrices entry by entry. This mismatch between how a model is expressed and how it is implemented made them difficult to verify, modify, and communicate with others [9]. *Algebraic modeling languages* such as AMPL [10], AIMMS [11] and GAMS [12] closed this gap by accepting problem descriptions that closely follow the math. For a more complete overview of the early history of modeling languages refer to [13].

More recently, modeling languages embedded in general-purpose programming languages have gained popularity, such as Pyomo [14,15] in Python, JuMP [16] in Julia, and CasADi [17] in C++. Other domain-specific modeling languages specialized for convex optimization [18] are built on *disciplined convex programming* (DCP), a set of composition rules under which the convexity of a problem can be verified automatically [3, 4]. These DCP-based languages include CVX [19] in MATLAB, CVXR [20] in R, CVXPY [1] in Python, and Convex.jl [21] in Julia.

Structure of the chapter The first two sections focus on CVXPY: it is the most actively maintained DCP implementation, and the work of this thesis aims to be integrated into it. §2.1 covers the *frontend* of canonicalization, which rewrites the user's functional form into a functional conic form of purely affine and quadratic expressions. §2.2 covers its *backend*, *matrix stuffing*, which assembles the problem data from this functional conic form.

Because the problem data assembled this way is exactly a sparse Jacobian of the canonicalized form, §2.3 reviews the methods available for computing sparse derivatives. §2.4 then revisits the same task from the side of other modeling systems: it presents the embedded modeling languages JuMP and CasADi as well as the algebraic language AMPL as instances of that machinery, and compares how each one extracts the problem data with the approach taken in CVXPY. Finally, §2.5 covers the sparse matrix formats used to store the problem data,

along with some useful algorithms for manipulating them.

2.1 Canonicalization

CVXPY is a DCP-based modeling language embedded in Python [1], which began in 2013 as an implementation of the modeling principles of CVX [19]. Three fundamental design decisions differed from CVX at the beginning. The first is modularity: a few core data structures are shared by separate components for verification, canonicalization, matrix stuffing, and solver interfaces. The second is the expression tree as the central data structure, which is formed automatically from user code through Python's operator overloading. The third is support for parameters: since canonicalization never looks at the values of constants, much of the compilation work can be reused when a problem is solved repeatedly.

These ideas matured with version 1.0, which redesigned canonicalization as a general reduction-based rewriting system [22]. The new design enabled the addition of other disciplined grammars [23–25], and the early parameter support grew into disciplined parameterized programming (DPP), a restricted grammar under which canonicalized programs can be updated efficiently [26].

The open source project has since grown through contributions from many other enthusiastic developers; a recent paper surveys the new canonicalization and modeling features introduced between versions 1.1 and 1.9 [27].

Functional form. A user most naturally expresses a CVXPY problem in *functional* form,

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & f_i(x;\theta) \le g_i(x;\theta), \quad i = 1,\dots,m_1, \\ & h_i(x;\theta) = j_i(x;\theta), \quad i = 1,\dots,m_2, \end{array} \tag{2.1}$$

where the optimization variables $x_1,\dots,x_k$ are collected into a single vector $x \in \mathbf{R}^n$, and likewise the parameters $\theta_1,\dots,\theta_l$ into $\theta \in \mathbf{R}^p$ [18]. We call (2.1) a problem *instance* when θ is fixed to a numerical value, and a problem *family* when θ is left symbolic. Internally, CVXPY represents each function in (2.1) as an expression tree.

Disciplined convex programming. When f_0 and all f_i are convex, all g_i are concave, and all h_i and j_i are affine, the functional form defines a convex optimization problem. To certify convexity, CVXPY uses the DCP composition rules to classify the curvature of each function. Once all functions are classified, a problem is *DCP-compliant* if the objective

and each constraint have curvatures compatible with a convex problem: a convex objective, convex $\leq$ concave inequalities, and affine equalities.

Otherwise, if a curvature is incompatible (*e.g.* f_0 is concave) or cannot be determined, CVXPY raises an error and terminates. Because these rules are only sufficient for convexity, a problem that is genuinely convex but not expressed in a recognized form may still be rejected.

Composition rules. The composition $\phi(f_1(x), f_2(x), \ldots, f_k(x))$ of an atom ϕ with subexpressions f_i is convex when ϕ is convex and, for each argument $i \in \{1, \ldots, k\}$, one of the following holds:

- ϕ is non-decreasing in argument i and f_i is convex, or
- ϕ is non-increasing in argument i and f_i is concave, or
- f_i is affine.

The same results hold for concave functions by swapping convex and concave. Affine expressions require ϕ and all of its arguments to be affine [3].

Functional conic form. Once a problem is verified to be DCP-compliant, the frontend of canonicalization rewrites its functional form (2.1) into an equivalent *functional conic form*

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & \ell_i(x;\theta) \in \mathcal{K}_i, \quad i = 1, \ldots, m, \end{array} \tag{2.2}$$

where f_0 is affine or quadratic, each ℓ_i is affine, and each $\mathcal{K}_i$ is one of the standard convex cones: the zero, nonnegative, second-order, power, exponential, and positive semidefinite cones.

Reductions. This rewriting is carried out as a sequence of *reductions*. Each reduction provides an `apply` step that converts a problem into an equivalent one, and a `retrieve` step that recovers a solution of the original problem from a solution of the rewritten one [22].

A first example is `FlipObjective`, which turns a maximization problem into a minimization one by flipping the sign of the objective; its `retrieve` step simply flips the sign back.

A second example, `CvxAttr2Constr`, handles variable attributes, which are properties declared on a variable that restrict its domain. These include nonnegativity (`nonneg`), positive

semidefiniteness (`PSD`), and explicit variable bounds, amongst others. When the target solver supports an attribute natively, CVXPY leaves it for the solver to enforce; otherwise the reduction creates a new variable that carries no attributes and adds explicit constraints that enforce the attribute. Its `retrieve` step then maps the reformulated solution back onto the original attributed variable.

Conic reformulation. The main reduction, `Dcp2Cone`, rewrites every nonlinear atom into conic constraints. The first step of the reduction is to rewrite the problem into *Smith form*, where each nonlinear atom is replaced by a new variable coupled with an equality constraint; every atom then acts only on affine arguments [28]. For a convex atom ϕ, the Smith form gives an auxiliary variable t with the equality constraint $\phi(x) = t$. The second step relaxes this equality to the inequality $\phi(x) \leq t$. The DCP rules allow a convex atom only where its value is minimized or upper-bounded, so the solver always prefers t as small as possible; the relaxation is therefore *lossless*.

The pair (x, t) then lies in the *epigraph* $\{(x, t) : \phi(x) \leq t\}$ of ϕ, so we often refer to t as an *epigraph variable*. A concave atom relaxes to $\phi(x) \geq t$ instead, giving a *hypograph variable*. The final step of the reduction expresses each epigraph (hypograph) inequality as the membership of an affine function of (x, t) in a convex cone [29].

Example of reformulations. A first example is the ℓ_2 norm atom $\|x\|_2$, whose epigraph $\|x\|_2 \leq t$ is exactly the second-order cone constraint $(t, x) \in \mathcal{K}_{\text{soc}}$. A second example is the nondifferentiable absolute value atom $|x|$, whose epigraph $|x| \leq t$ can be written smoothly as the pair of affine inequalities $-t \leq x \leq t$. A final example is the sum of the k largest entries of x, which expands into a small LP: the epigraph sum_largest$(x, k) \leq t$ holds if and only if there exist a scalar q and a vector u such that

$$q\mathbf{1} + u \geq x, \qquad u \geq 0, \qquad kq + \mathbf{1}^\top u \leq t. \tag{2.3}$$

Here q acts as a threshold and each u_i measures how far the entry x_i rises above it, so that the sum of the k largest entries is bounded by $kq + \mathbf{1}^\top u$ [18].

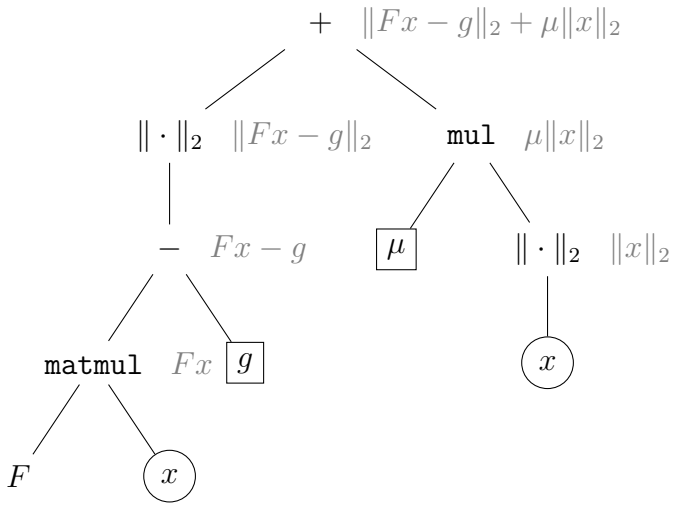
Example of canonicalization. To show a concrete example of the frontend of canonicalization, we consider a simple fitting with regularization problem

$$\begin{array}{ll} \text{minimize} & \|Fx - g\|_2 + \mu\|x\|_2 \\ \text{subject to} & x \succeq 0 \end{array} \tag{2.4}$$

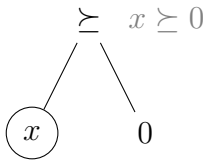
with variable $x \in \mathbf{R}^n$, fixed problem data $F \in \mathbf{R}^{m \times n}$, and parameters $g \in \mathbf{R}^m$ and $\mu \geq 0$. The problem formulation in CVXPY code is

```
import cvxpy as cp
import numpy as np
F = np.random.randn(m, n)
x = cp.Variable(n)
g = cp.Parameter(m)
mu = cp.Parameter(nonneg=True)
objective = cp.Minimize(cp.norm(F @ x - g) + mu * cp.norm(x))
prob = cp.Problem(objective, [x >= 0])
```

where we also use the numerical computation library `NumPy` [30] for the data arrays. The expression tree representation of the problem is shown in 2.1.



(a) Expression tree of the objective



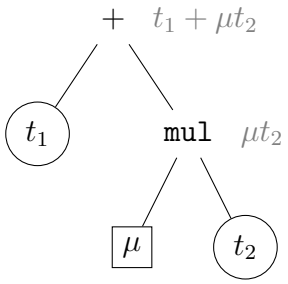
(b) Expression tree of the constraints

Figure 2.1 *Before canonicalization.* CVXPY represents the DCP problem (2.4) as these expression trees. Variables are labeled with a circle, and parameters with a square.

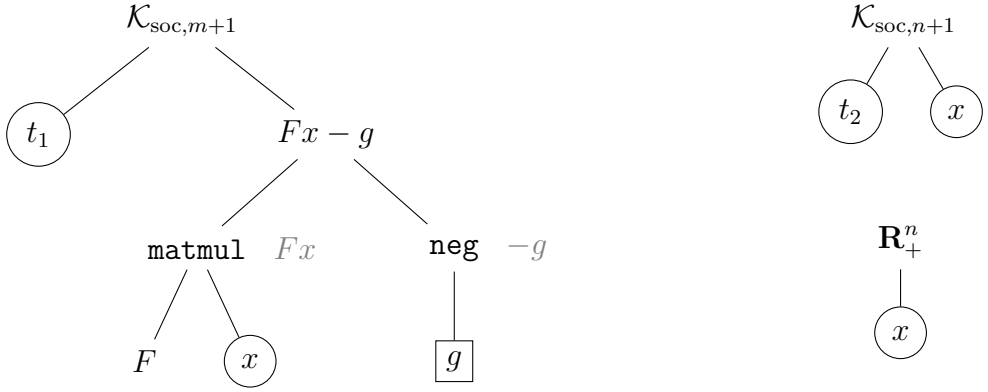
Conic form expression tree. After applying the reductions, the problem has been *canonicalized* to

$$\begin{array}{ll} \text{minimize} & t_1 + \mu t_2 \\ \text{subject to} & x \in \mathbf{R}^n_+ \\ & (t_1, Fx - g) \in \mathcal{K}_{\text{soc},m+1} \\ & (t_2, x) \in \mathcal{K}_{\text{soc},n+1} \end{array} \tag{2.5}$$

where t_1 and t_2 are new epigraph variables, and the cone $\mathcal{K}$ from 1.1 is the product $\mathbf{R}^n_+ \times \mathcal{K}_{\text{soc},m+1} \times \mathcal{K}_{\text{soc},n+1}$. Problem (2.5) is now in the functional conic form introduced above: every constraint is a cone membership, and the objective and constraint expressions are affine. Comparing the DCP trees of 2.1 with the conic-form trees of 2.2 shows the effect of canonicalization: the nonlinear norm atoms have been replaced by epigraph variables, leaving affine expression trees in the variables. The three rows of (2.5) remain three separate cone constraints, each carrying the affine expressions that must lie in its cone; 2.2 accordingly shows one expression tree per constraint, rather than a single combined tree.



(a) Expression tree of the objective



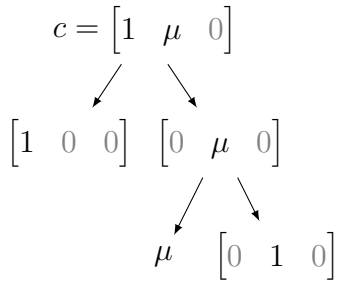
(b) One expression tree per constraint

Figure 2.2 *After canonicalization.* CVXPY represents the conic form (2.5) as these expression trees: the nonlinear norm atoms have become epigraph variables, leaving the objective and the three separate cone constraints affine. Variables are deduplicated and ..

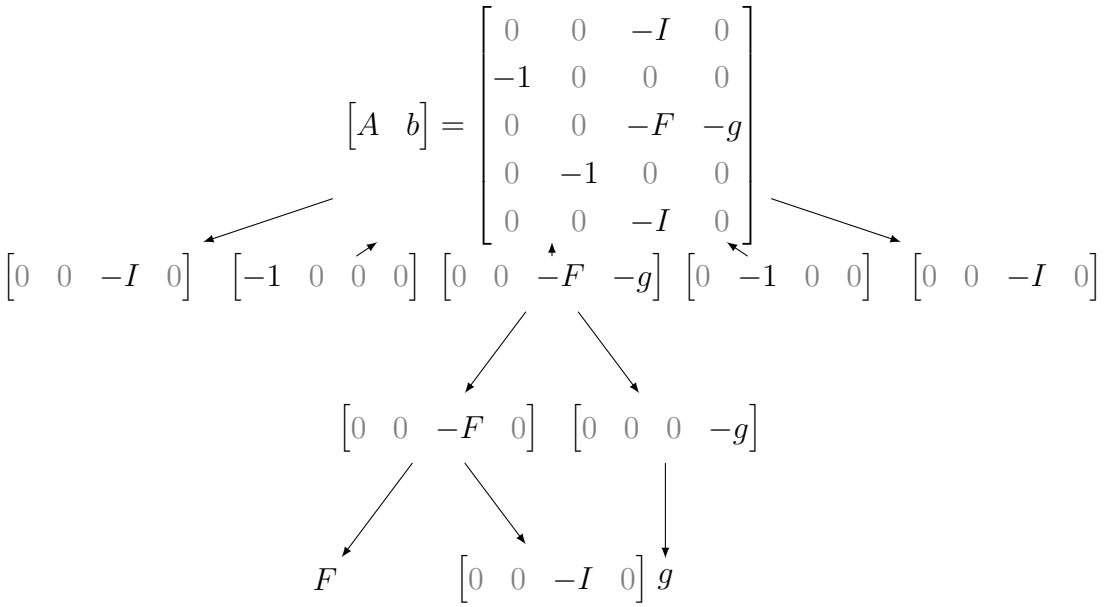
References. Michael Grant's papers and PhD thesis on CVX and DCP are the standard references on verifying convexity and canonicalization [3, 4, 29]. The paper by Chu et al. [31] gives another perspective of canonicalization through the *relaxed smith form*, it was used extensively to learn the concept initially. The paper by Agrawal et al. [22] on rewriting systems is a great reference for understanding reductions and the core concepts behind CVXPY. Fi-

nally, the Mosek modeling cookbook is a complementary reference for epigraph reformulations and general modeling best practices [32].

2.2 Matrix stuffing



(a) Matrix stuffing of the objective

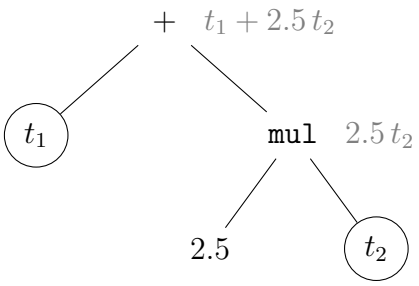


(b) Matrix Stuffing tree

Figure 2.3 Matrix stuffing of problem (2.4) by a recursive walk of the conic-form expression trees of 2.2: each node's matrix is obtained by applying a linear operation to the matrices of its children.

$$Q = \begin{array}{c} \\ t_1 \\ t_2 \\ x \\ d \end{array} \begin{array}{c} \begin{array}{ccc} g^T & \mu & 1 \end{array} \\ \begin{bmatrix} 0 \cdots 0 & 0 & 1 \\ 0 \cdots 0 & 1 & 0 \\ 0 \cdots 0 & 0 & 0 \\ 0 \cdots 0 & 0 & 0 \end{bmatrix} \end{array} \qquad \begin{bmatrix} c \\ d \end{bmatrix} = Q\,\tilde{\theta} = \begin{bmatrix} 1 \\ \mu \\ 0 \\ 0 \end{bmatrix}$$

(a) DPP path: the objective parameter map Q is formed and cached; each re-solve recovers the objective data (c, d) by multiplying Q with the extended parameter vector $\tilde{\theta} = (g, \mu, 1)$.



$$\Rightarrow \quad c = \begin{bmatrix} 1 & 2.5 & 0 \end{bmatrix}$$

(b) `ignore_dpp` path: `EvalParams` evaluates the parameters in the functional form, before canonicalization, so the canonicalized objective tree carries the constant 2.5 in place of the parameter leaf μ that the general path keeps (2.2). Canonicalizing this constant-data tree yields the numeric c directly, but the whole chain re-runs whenever a parameter changes.

Figure 2.4 The two compilation paths on the objective of problem (2.4): the DPP path caches the parameter map Q once and re-solves with a matrix-vector product, while the `ignore_dpp` path evaluates the parameters up front and re-canonicalizes from scratch.

Whichever path CVXPY takes, matrix stuffing ends in the same place: a set of constant, sparse problem-data matrices read off from the canonicalized affine form. These matrices are nothing other than the sparse Jacobian of that form with respect to the variables, and every modeling language must produce them in one guise or another. We therefore review in §2.3 how sparse derivatives are computed, before returning in §2.4 to how each system carries out this extraction.

References. The canonicalization backends are largely undocumented in the literature; the CVXPY and `cvxcore` source code are the primary references for this section, together with the rewriting system paper [22]. The DPP ruleset and the cached parameter maps were introduced in the paper on differentiable convex optimization layers [26].

2.3 Computing derivatives

Finding the rate of change, or *derivative*, of a function is a fundamental problem in many fields of science and engineering. Although computing derivatives may seem straightforward at first, there are many approaches, each with its own strengths and trade-offs [48]. These approaches differ in four basic choices: whether the function is treated as a *scalar* or *vectorized* expression, whether derivatives are propagated *forward* or in *reverse*, whether the Jacobian or Hessian is assembled *directly* or accessed through an *oracle* of matrix–vector products, and whether the result is *sparse* or *dense*.

The problem data for an optimization model is often large and sparse, so we place particular emphasis on computing sparse derivatives in this section. The standard approach in the literature is scalar and oracle-based differentiation [41], where graph coloring is used to recover a sparse matrix in far fewer oracle evaluations. This thesis instead proposes to compute these sparse derivatives directly on a vectorized graph whose sparsity pattern follows from the structure of the operations, so no coloring is needed.

The remainder of the section is structured as follows: we begin with a brief survey of the standard numerical and symbolic methods, then cover *algorithmic differentiation* (AD) from the perspective of these four choices. Finally, we gather a range of relevant works, organizing them in a table according to their particular approach to differentiation.

Numerical differentiation. Other than computing derivatives by hand, the simplest method is to use *numerical differentiation*, which approximates the derivative of a function at a point by evaluating the function at nearby points using a *finite difference* formula similar to either

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \tag{2.6}$$

or

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \tag{2.7}$$

where h is a small step size. Although this approach is straightforward to implement and can achieve reasonable accuracy, there are important downsides to consider. First, it requires a linear number of function evaluations with respect to the number of input variables, which is expensive for high-dimensional functions. Second, it can easily lead to *truncation errors* (when h is too large) or *rounding errors* (when h is too small) in the computed derivative if the step size is not carefully chosen [42]. Despite these downsides, numerical differentiation is still widely used in practice, especially for verifying the correctness of derivative computations (also known as a *derivative checker*) [43].

Symbolic differentiation. Mathematical expressions can be manipulated automatically using simple derivative rules of calculus, such as the product and sum rule. This *symbolic* approach to differentiation is used in computer algebra systems such as Mathematica [44] and Maple [45] as well as the SymPy library in Python [46]. Unlike its numerical counterpart, symbolic differentiation tools compute derivatives as closed-form expressions, which can be simplified and evaluated at any arbitrary point. These closed-form expressions are particularly useful for deriving analytic solutions of optimization problems, and can be further used in mathematical proofs and analyses.

However, a naive implementation of symbolic differentiation can easily lead to exponentially large symbolic expressions, which is impractical and inefficient for many problems [41]. Interestingly, when intermediate expressions are stored as *common subexpressions*, and the expression is represented with a *directed acyclic graph* (DAG) instead of a tree, symbolic differentiation performs the same set of operations as *reverse mode differentiation* [47].

Forward mode AD. The method of propagating tangents forward through the computational graph is known as *forward mode* AD. A single forward pass simultaneously computes the value and derivative of every intermediate variable with respect to a single input variable, giving the sensitivity of all outputs to that input. The cost of a forward pass is a small constant factor of the cost of evaluating the original function, since each elementary operation is augmented with at most a few additional arithmetic operations for the derivatives. However, one downside of forward mode AD is that the number of forward passes scales linearly with the number of input variables, which is impractical for certain functions (which have many input variables and very few output variables).

Reverse mode AD. In contrast to forward mode, *reverse mode* AD propagates adjoints backward through the computational graph. It first requires a forward pass to compute and store the value of every intermediate variable, and it then performs a reverse pass to compute the sensitivities of a single output variable with respect to all input variables. The cost of a reverse pass is also a small constant factor of the cost of evaluating the original function, but the number of passes required now scales linearly with the number of output variables rather than the number of input variables. This makes reverse mode AD particularly efficient for computing gradients of scalar-valued functions, making it the default method for modern machine learning libraries such as Jax [49], Pytorch [50], and Tensorflow [51].

From scalar gradients to sparse problem data. The methods above were developed for the dominant use of AD: computing a dense gradient of a scalar-valued function. Extract-

ing optimization problem data instead calls for a large, sparse Jacobian of a vector-valued function, and reaching it takes two independent shifts in how derivatives are computed, which we take in turn.

Scalar versus vectorized graphs. The first shift concerns whether the computational graph is *scalar* or *vectorized*. In a *scalar* graph each node is a single scalar operation, so a matrix product Ax with $A \in \mathbf{R}^{m\times n}$ expands into roughly mn multiplication and addition nodes, and the local derivative attached to each edge is a single number. In a *vectorized* graph the same product is one node, and its local derivative is no longer a scalar but a *structured linear operator*. In the matrix differential calculus of Magnus and Neudecker [52] these operators take a compact form built from Kronecker products, diagonal matrices, and selection matrices; the differential of a matrix product $C = AB$, for instance, is $\mathrm{d}\operatorname{vec} C = (B^\top \otimes I)\,\mathrm{d}\operatorname{vec} A + (I \otimes A)\,\mathrm{d}\operatorname{vec} B$. Differentiation in a vectorized graph is therefore the *composition of linear operators* rather than the accumulation of scalar chain-rule products, and each local operator carries the macro-structure of its operation instead of shattering it into individual entries. This choice governs both the size of the graph—$O(mn)$ scalar nodes against a handful of array nodes—and the shape of the local derivatives. Scalar graphs underlie classical AD tools such as ADOL-C [53] and CppAD [54], whereas a recent line of work computes derivatives in vectorized, sparse, forward-mode form across several domains: optimal control [55], statistical estimation [56], and sparse-tensor differentiable programming [57].

Dense versus sparse derivatives. The second shift concerns the *data format* in which the derivative is carried. A dense representation stores every entry of the Jacobian, including the structural zeros; a *sparse* representation stores and propagates only the nonzeros. For the matrices that arise in optimization this distinction is decisive: a constraint matrix with millions of entries may have only a few nonzeros per row, so forming it densely is not merely wasteful but often infeasible. Keeping the derivative sparse means representing it in one of the compressed formats discussed in §2.5 and arranging every elementary operation to act on the nonzeros alone. The two shifts are independent: a graph may be scalar or vectorized regardless of whether its derivatives are dense or sparse, and it is their *combination*—a vectorized graph carrying sparse derivatives—that the data-extraction problem of this thesis calls for. What remains is the question of *how* the sparse matrix is actually produced, and here the literature divides into two routes.

Sparse Jacobians by coloring. The first route never forms the Jacobian explicitly. A single AD pass does not return the matrix; what it returns depends on what is propagated through the graph. Propagating one seed vector gives a *matrix-free*, or *oracle*, primitive: a forward pass returns the *Jacobian–vector product* Jv and a reverse pass the *vector–Jacobian product* $w^\top J$, each a projection of J that costs about one evaluation of the function [48]. To recover the matrix itself one repeats the pass over a basis of seed directions—n of them in the worst case. When J is sparse this basis can be compressed: two columns that are never both nonzero in the same row are *structurally orthogonal* and can be recovered together from a single combined seed, so minimizing the number of seeds becomes a graph-coloring problem on a graph built from the sparsity pattern. This idea was introduced by Curtis, Powell, and Reid [60] and cast as a coloring problem by Coleman and Moré [61]; the survey of Gebremedhin, Manne, and Pothen [62] unifies its many variants. A further ingredient is *sparsity detection*—recovering the pattern itself when it is not given in advance—which modern operator-overloading systems perform automatically and which has become the dominant cost of the pipeline [64]. Seen this way, coloring is a means of *recovering* a structure that the matrix-free route has discarded: because the matrix is never formed, its pattern must be rediscovered and then probed through the oracle. Vectorization alone does not remove this cost—CasADi's matrix-valued `MX` graph is vectorized, yet to extract the sparse matrices H and A it still detects their pattern and colors it to recover the entries from a few seeded sweeps (§2.4).

Direct sparse accumulation. The second route keeps the derivative sparse and assembles the matrix directly. Each elementary operation contributes a local Jacobian, and composing these—in the language of Griewank and Walther [41], *eliminating* the intermediate vertices of the graph—yields J itself, with no seed directions. Assembling the matrix this way does not require storing the Jacobian of every intermediate variable: in a bottom-up pass each local contribution is folded into its successor and then released, so only the final matrix and the operands of the step in progress need be held at once. Forward and reverse mode are simply two orderings in which the vertices are eliminated, and choosing the order that minimizes the arithmetic is the *optimal Jacobian accumulation* problem, which is NP-complete on a general graph [58] and remains so even structurally, under vertex elimination [59]. This combinatorial difficulty, however, is specific to dense, scalar graphs: on a vectorized graph the local Jacobians are structured sparse operators, the graph itself is small, and a single bottom-up pass forms the matrix directly and cheaply, so neither a seed basis nor a search over elimination orders is required. The idea of carrying the derivative as a sparse object is not new: the SparsLinC library of ADIFOR already propagated derivative objects as sparse

vectors with *dynamic sparsity*, exploiting structure as the computation proceeded rather than recovering it afterwards by coloring [65], and collected matrix-derivative rules give the forward and reverse sensitivities of products, inverses, and factorizations at the level of whole array operations [67]. The recent vectorized, forward, sparse methods [55–57] and the coefficient extractor of this thesis belong to this direct lineage; what they share is an array-level graph with analytically known per-operation sparsity, so that the full sparse matrix is composed in a single pass without ever forming a seed basis.

Sparse Hessians. The same two routes reappear for second derivatives. The matrix-free route assembles the Hessian as the Jacobian of the gradient and exploits symmetry through *star* and acyclic colorings, which recover a pair of mirror entries H_{ij} and H_{ji} from a single seed [63]; this is how CasADi forms the objective Hessian P, by a forward-over-reverse sweep over a star-colored seed matrix (§2.4). The direct route instead builds the sparse Hessian on the graph itself: the edge-pushing algorithm accumulates it together with its values in a single pass, bypassing detection, coloring, and compression [66]; at a coarser grain, AMPL reaches the same end by exploiting the partial separability of a scalar model [37]. The coefficient extractor of this thesis follows the direct route for the objective Hessian of a quadratic program, assembling it as a structured sum of sparse blocks whose pattern is fixed in advance, with no symmetric coloring required.

Differentiation packages. A wide range of AD software exists. For `C` and `C++`, ADOL-C [53] and CppAD [54] implement scalar operator overloading, recording an evaluation trace that is then replayed in either mode. Source-to-source tools instead transform the program at compile time: Tapenade [68] for `Fortran` and `C`, and Enzyme [69] at the level of the `LLVM` intermediate representation, which lets it differentiate code written in any compiled language. In the deep-learning setting, the reverse mode of Jax [49], Pytorch [50], and Tensorflow [51] operates on vectorized graphs but returns dense gradients; sparse autograd is only beginning to appear there, for instance differentiating through sparse linear solves [70]. The optimization-oriented frameworks AMPL [37] and CasADi [17] are examined in detail in §2.4.

Summary. The methods surveyed here can be characterized by four choices: the *expression type* (scalar or vectorized), the *mode* (forward, reverse, or both), whether the Jacobian is formed *directly* or matrix-free through an *oracle*, and the *data format* of the derivatives (sparse or dense). Table 2.1 places the main tools and recent works according to these choices. The coefficient extractor developed in this thesis occupies the cell of vectorized,

forward, direct, and sparse differentiation, which it shares with the recent works of Zou and Jiang [55], Kwok et al. [56], and ∇SD [57]. What distinguishes the present work is the program being differentiated: because it is the affine tree produced by canonicalization, the sparsity pattern is known per atom and the Jacobian recovered is exactly the constant conic problem data, so graph coloring never enters the picture. This connection between coefficient extraction and differentiation, obvious in hindsight, is the starting point for the methodology developed in this thesis.

Table 2.1 Differentiation methods classified by the four choices of this section: the *expression type* (scalar or vectorized), the *mode* (forward, reverse, or both), the *assembly* of the Jacobian (formed *directly* by composing local Jacobians, or matrix-free through an *oracle* that returns only matrix–vector products and is reconstructed by seeded sweeps), and the *data format* (sparse or dense). The vectorized, forward, direct, sparse methods (lower block) need no graph coloring, unlike the matrix-free methods above them.

Method / work	**Expression type**	**Mode**	**Assembly**	**Data format**
ADOL-C, CppAD [53, 54]	scalar	both	oracle	dense
Tapenade, Enzyme [68, 69]	scalar	both	oracle	dense
CasADi [17]	scalar / vectorized	both	oracle	sparse
Jax [49]	vectorized	both	oracle	dense
Pytorch [50]	vectorized	reverse	oracle	dense
`torch-sla` [70]	vectorized	reverse	oracle	sparse
CPR, Coleman–Moré [60, 61]	scalar	forward	oracle	sparse
Hill–Dalle [64]	scalar	both	oracle	sparse
Vector forward mode [71]	scalar	forward	oracle	dense
Zou and Jiang [55]	vectorized	forward	direct	sparse
Kwok et al. [56]	vectorized	forward	direct	sparse
∇SD [57]	vectorized	forward	direct	sparse
This thesis	**vectorized**	**forward**	**direct**	**sparse**

References. The book by Griewank and Walther [41] is a comprehensive reference on the theory and practice of scalar automatic differentiation, including the graph-elimination view of Jacobian accumulation. The book by Naumann [72] is another great reference that focuses on the implementation of AD in software, with a particular emphasis on the use of operator overloading and source code transformation techniques. The survey by Baydin et al. [48] is an excellent and relatively up-to-date overview of the various methods for automatic differentiation in machine learning. For the combinatorial side of sparse derivative computation, the survey of Gebremedhin et al. [62] is the standard entry point, and the illustrated guide of Hill and Dalle [73] gives an accessible modern tutorial on the detection–coloring–compression pipeline.

2.4 Coefficient extraction

Matrix stuffing is CVXPY's particular answer to a task that every modeling language must perform: converting a symbolic model into the numerical problem data consumed by a solver. In this section, we review how three other modeling systems approach this *coefficient extraction* task and contrast their designs with CVXPY.

`JuMP.` JuMP is an algebraic modeling language embedded in Julia whose development started in 2012 at the MIT Operations Research Center by Iain Dunning, Joey Huchette, and Miles Lubin, with a first public release in 2013 [16,33]. It is now maintained by a team of core contributors, with Oscar Dowson as its most active maintainer. Unlike CVXPY, JuMP performs no convexity verification; it belongs to the derivative-based modeling paradigm, where the user is responsible for the properties of the formulation. Coefficient extraction in JuMP is *eager*: the `@constraint` and `@objective` macros rewrite the user's code at parse time into loops that build each affine expression as a sparse list of terms (an ordered dictionary mapping variables to coefficients). This macro rewriting avoids the asymptotic blowup of naive operator overloading, where summing d terms with repeated calls to `+` copies partial sums of growing size. As a consequence, the problem data is scalarized one constraint at a time while the model is being built, whereas CVXPY retains whole expression trees until solve time.

`MathOptInterface.` JuMP communicates with solvers through MathOptInterface (MOI), an abstraction layer whose standard form is to minimize $f_0(x)$ subject to constraints $f_i(x) \in S_i$, where the functions f_i and sets S_i are drawn from predefined lists [34]. There is deliberately no single canonical form: each solver advertises the function-set pairs it supports natively, and unsupported constraints are rewritten by *bridges*, local transformations selected per constraint type by a shortest-path search over a hypergraph of possible reformulations. Bridges play the same role as CVXPY's reductions, but they are chosen just-in-time as constraints are added to the model, rather than applied as a global pass over the whole problem. The constraint matrix only materializes inside the solver wrapper. When the model is copied to the solver, MOI assembles the affine terms of all constraints into a single sparse matrix in compressed sparse column (CSC) format through a two-pass, sort-free procedure: a first pass counts the nonzeros falling in each column and a prefix sum turns these counts into the column pointers, sizing the row and value arrays to exactly the number of nonzeros; a second pass scatters each term into its slot using a per-column cursor. The column indices are stored either one-based, for an allocation-free view as a native sparse matrix, or zero-based, so the

arrays can be handed directly to a C solver. This assembly is the same coordinate-to-CSC computation of §2.5, but it is carried out for a single fully numeric instance: the standard form $f_i(x) \in S_i$ admits no symbolic parameters, so re-solving with changed data requires rebuilding the affine terms or modifying individual coefficients in place and re-copying to the solver. There is no precompiled map from a parameter vector to the problem data of the kind CVXPY caches (1.3); even `ParametricOptInterface.jl`, which adds parameters to the JuMP ecosystem, substitutes their fixed values and re-optimizes rather than applying a reusable sparse parameter-to-data map.

`AMPL.` AMPL is one of the original standalone algebraic modeling languages, developed by Robert Fourer, David Gay, and Brian Kernighan at Bell Laboratories and Northwestern University [10] to replace the error-prone matrix generators that practitioners previously wrote by hand [9]. An AMPL model is purely symbolic, with variables and constraints indexed over sets; the AMPL translator instantiates the model with data, applies a presolve, and writes the problem instance to a `.nl` file that is read by the solver [35]. The file splits each constraint into a linear part, stored as sparse coefficient lists along with the Jacobian sparsity pattern, and a nonlinear part, stored as an expression graph in Polish prefix notation. Inside the solver process, the AMPL Solver Library (ASL) reads the file, exposes the linear coefficients (in CSC format if the solver requests it), and evaluates the nonlinear Jacobian entries with reverse mode algorithmic differentiation (AD) on the expression graphs [36, 37].

`CasADi.` CasADi is a software framework for nonlinear optimization and optimal control built around algorithmic differentiation [17]. Expressions are stored in two graph types, scalar-valued `SX` nodes and matrix-valued `MX` nodes, and every matrix in the framework is sparse in CSC format. CasADi can also target nonlinear solvers, to which it hands AD-generated derivative callbacks rather than any explicit coefficient matrix; the case relevant here, however, is its interface to conic solvers such as Clarabel. On that path the `conic` front-end differentiates the symbolic objective and constraints to identify the problem data — the matrices H and A together with their sparsity patterns — which it then forwards to the solver. The sparsity patterns are detected by propagating bitvectors through the graph, and their numerical values are recovered in a few AD sweeps using the graph coloring techniques covered in 2.3. Second-order-cone constraints are accepted on this same path through a reformulation helper, but CasADi stops short of the disciplined-convex canonicalization that lets CVXPY target general conic solvers automatically. Crucially, this differentiation step presupposes a problem already cast in the solver's standard form: AD recovers H and A from a smooth objective and smooth constraints, but it cannot turn a nonsmooth convex atom —

a norm, a maximum, an eigenvalue — into the auxiliary variables and cone constraints that represent it. Supplying that epigraph reformulation is left to the user, whereas it is precisely the work that CVXPY's canonicalization performs automatically.

Comparison. As established in §2.3, preparing the problem data amounts to computing the sparse Jacobian of the constraint functions with respect to the variables; each of these systems is therefore a particular point in that landscape. JuMP accumulates it eagerly, one scalarized constraint at a time; AMPL writes it to a file and defers the nonlinear entries to AD inside the solver; CasADi extracts it from the symbolic graph by AD; and CVXPY stuffs it from canonicalized affine trees, with the additional map from parameters to problem data. Where they differ most sharply is one level up, in the front-end they expose: JuMP, AMPL, and CasADi are algebraic, requiring the problem to be stated in a form the target solver already accepts, whereas CVXPY alone infers convexity from a disciplined grammar and reduces it to conic form on the user's behalf. Table 2.2 lays out the four systems along these two axes — the front-end paradigm and the mechanism by which problem data is materialized.

Table 2.2 Comparison of problem-data extraction methods across modeling languages.

	JuMP	**AMPL**	**CasADi**	**CVXPY**
Front-end paradigm	Algebraic modeling	Algebraic, file-based	Symbolic graphs + AD	Disciplined convex (DCP)
Problem classes	LP, MILP, QP, conic, NLP	LP, MILP, NLP (conic via recent drivers)	NLP, QP / conic (QP-shaped)	Convex: LP, QP, SOCP, SDP, EXP
Path to conic solvers	Native cones + bridges (MOI)	Solver-side, recent drivers	`soc` helper on the QP path	Automatic DCP reduction
Problem-data mechanism	Eager, per scalar constraint	`.nl` file; nonlinear via solver AD	Explicit H, A by AD on the graph	Affine-tree stuffing + parameter map

References. The MathOptInterface paper [34] is the best reference on the design of JuMP's solver abstraction layer and its bridge system. David Gay's technical reports [35, 36] remain the authoritative description of the `.nl` format and the AMPL solver interface. Finally, the CasADi paper [17] is a self-contained reference covering its expression graphs, AD implementation, and solver interfaces.

2.5 Sparse matrices

Matrices that contain many number of zero entries are known as *sparse matrices*. The simplest sparse matrix data structure is the *coordinate* or *triplet* format. It consists of two integer index arrays `rows` and `cols` and a floating point value array `vals` of length equal to the number of *nonzero entries* in the matrix.

For example, the matrix

$$\begin{bmatrix} 0 & 0 & 3.2 & 0 \\ 4.1 & 1.8 & 0 & 2.2 \\ 0 & 2.8 & 0 & 3.4 \end{bmatrix} \tag{2.8}$$

can be represented in coordinate format as

```
int    rows[] = {0,  1,   1,   1,   2,   2};
int    cols[] = {2,  0,   1,   3,   1,   3};
double vals[] = {3.2, 4.1, 1.8, 2.2, 2.8, 3.4};
```

where we use zero-based indexing and array declaration notation from the C programming language [74]. The coordinate format is the easiest to understand and form, but it is not effective for most operations on sparse matrices; the *compressed sparse row* or CSR format is commonly used instead.

Compressed sparse row format. The CSR format differs from the coordinate format in that it replaces the `rows` array with a row pointer array `idxptr` of length equal to the number of rows plus one. The row pointer array indicates the starting index of each row in the `cols` arrays. For example, the matrix in (2.8) can be represented in CSR format as

```
int  idxptr[] = {0,  1,              4,        6};
int   cols[] = {2,  0,   1,   3,   1,   3};
double vals[] = {3.2, 4.1, 1.8, 2.2, 2.8, 3.4};
```

where we deliberately format `idxptr` to align with the start of each row in the `cols` arrays. The benefits of the CSR format are more than a reduced size of the `idxptr` array compared to the `rows` array. In the CSR format, accessing a row takes constant time and is very *cache-friendly* since the nonzero entries are stored contiguously in memory. For this reason, the CSR format is the default for *iterative methods* to solve sparse linear systems [75].

Compressed sparse column format. However, one downside of the CSR format is that accessing a column is much more difficult; one would need to look at every row to find the nonzero entries in a column. An alternative format is the *compressed sparse column* or CSC

format, which swaps the role of the `idxptr` array to refer to column indices instead. In contrast to iterative methods, the CSC format is the preferred format for *direct methods* to solve sparse linear systems [76].

Transpose.

Which format to use?

Other formats. The coordinate, CSR and CSC formats are the most widely used sparse matrix formats, but many others exist for different specialized use cases. The scientific Python sparse library `scipy.sparse` [77] allows users to form other types of sparse matrices such as the list of lists (LIL) matrix, the dictionary of keys (DOK) matrix, and the block-sparse row (BSR) matrix.

Sparse matrix struct in C. A CSR matrix struct in C looks like this.

```
typedef struct CSR_Matrix {
    int *p;      /* row pointers, size m+1 */
    int *i;      /* column indices, size nnz */
    double *x;   /* values, size nnz       */
    int m, n, nnz;
} CSR_Matrix;
```

References. Although the work in this thesis does not necessarily revolve around solving sparse linear systems, a lot of techniques for manipulating sparse matrices we use were inspired by the literature. In particular, the book by Davis [76] was used extensively throughout the development of our methods. The book by Duff [78] is another great reference to learn more about the history and development of sparse matrix algorithms. Another reference is the review article by the "gang of four" (Gill, Murray, Saunders and Wright) from the Systems Optimization Laboratory at Stanford University [79] which provides a high-level overview of various applications of sparse matrix methods in optimization.

CHAPTER 3 A lightweight canonicalization engine

This chapter presents a minimal implementation of a subset of affine atoms in the `SparseDiffEngine` package, and shows how they compute the sparse constant Jacobian of a set of affine expressions. We start from a *canonicalized* problem in standard form, produced by the canonicalization of §2.1. The goal is the same as in §2.2: form the problem data. §3.2 describes the data structures, §3.3 the atom implementations, and §3.6 the problem class and its CVXPY interface. We assume the sparse-matrix background of §2.5.

3.1 Keeping parameters, splitting the walk

Matrix stuffing already walked the objective tree of our running example. Figure 2.3 annotated each node of the canonicalized objective $t_1 + \mu t_2$ of (2.5) with the row of c it produces, ending in $c = \begin{bmatrix} 1 & \mu & 0 \end{bmatrix}$ at the root. The engine walks the very same tree of 2.2, but it differs from matrix stuffing in two ways.

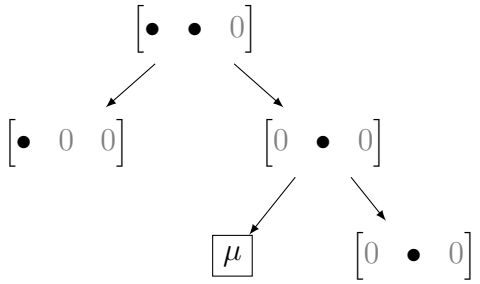
First, it keeps the parameter. The `ignore_dpp` path of §2.2 begins with `EvalParams`, which deletes the parameter leaf μ and replaces it by a constant holding its current value (2.4b); the walk that follows is parameter-free and must be redone from scratch whenever μ changes. We instead leave μ in place as a leaf node, so the tree that produces c is fixed once and reused across solves.

Second, it splits the single walk into two passes over that fixed tree, shown in 3.1. A *structural* pass fixes the sparsity pattern of every node's Jacobian from the patterns of its children, independent of the value of μ (3.1a). A *value* pass then fills the numbers into that fixed pattern, reading μ's current value at the leaf (3.1b). The structural pass runs once per problem family; only the value pass repeats on each solve, whereas matrix stuffing fuses the two and recomputes both on every change of μ. These passes are the `jacobian_init` and `eval_jacobian` functions that the rest of the chapter develops atom by atom.

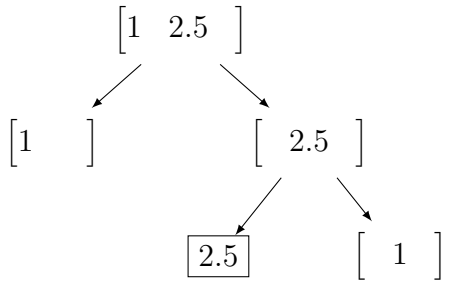
On this objective the saving is admittedly hard to see: the structural pass over $t_1 + \mu t_2$ is trivial, and one could fairly object that re-evaluating so small a tree from scratch costs nothing. The split earns its keep where the structural work is both expensive and shared across solves. The clearest case in our running example is the constraint tree of 2.3, which contains the product Fx (3.2). The Jacobian of Fx is a sparse matrix–matrix product handled by the matrix multiplication atom of §3.3, and its $-F$ block has a sparsity *pattern* that is the costly part to derive and is independent of any parameter value. Because the parameter

g enters the same constraint, the `ignore_dpp` path re-derives this block from scratch each time g changes, while the engine fixes its pattern once in the structural pass (3.2a) and only refills values (3.2b). The per-solve cost this removes, negligible for the scalar objective but substantial for problems with large sparse products solved repeatedly, is quantified in §4.

Like `cvxcore`, which mirrors the Python expression trees into equivalent C++ objects, the engine mirrors each CVXPY atom onto a C counterpart and rebuilds the canonicalized trees as the C `expr` trees of §3.2; both passes then run entirely in C.



(a) Structural pass: a bullet marks a structural nonzero.



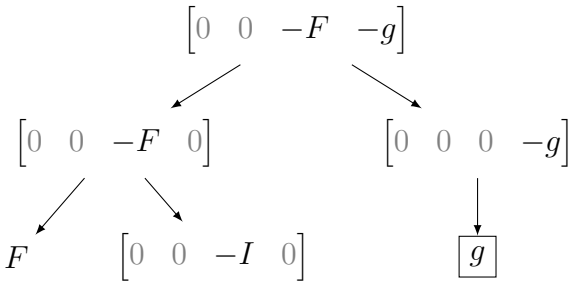
(b) Value pass: entries refilled here for $\mu = 2.5$.

Figure 3.1 The engine's two passes over the objective tree of 2.2, with the parameter leaf μ kept across both.

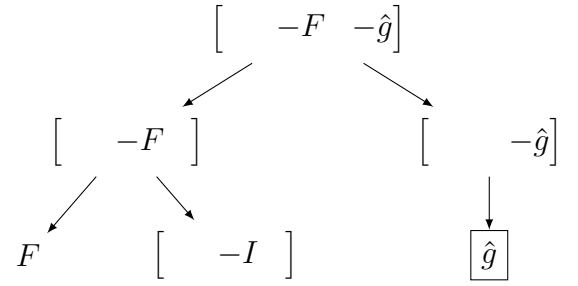
3.2 Fundamental data structures

The main data structure in the engine is the expression struct.

```
typedef struct expr {
    int   d1, d2, size; /* shape (d1 x d2) and size = d1*d2 */
    int   n_vars;       /* total number of variables     */
    int   var_id;       /* offset into u, or NOT_A_VARIABLE */
```



(a) Structural pass: the matmul block $-F$ inherits the pattern of F and is independent of g.



(b) Value pass: only the values are written into that pattern, here for $g = \hat{g}$. Between solves only the parameter block $-\hat{g}$ changes; the matmul block $-F$ is untouched.

Figure 3.2 The two passes on the $Fx - g$ subtree of the constraint tree of 2.3. The constant F enters through the matmul atom, whose block $-F$ has a sparsity pattern fixed once; across solves only the parameter g changes.

```
    struct expr *left;   /* left child (NULL for leaves)  */
    struct expr *right;  /* right child (NULL for leaves) */

    double    *value;    /* forward-pass output, size entries */
    CSR_Matrix *jacobian; /* size x n_vars sparse Jacobian */

    forward_fn     forward;
    jacobian_init_fn jacobian_init_impl;
    eval_jacobian_fn eval_jacobian;
} expr;
```

Each node carries a `value`, the output of its forward pass, and a sparse Jacobian in CSR form. Each node also implements three functions: `forward`, `jacobian_init`, and `eval_jacobian`.

Leaf nodes. The leaves are variables and parameters.

3.3 Atom implementations

An atom takes one or two expression operands and produces an expression. Each atom implements the three functions: `forward` computes the value, `jacobian_init` fixes the Jacobian's sparsity pattern once, and `eval_jacobian` fills its values on each solve. We give the full C implementation of one atom, `add`, to establish the pattern, and describe the rest in prose, each anchored by its CVXPY expression. Throughout, `x` and `y` denote variables, `a` and `b` parameters, and `F` a constant matrix, all of compatible shape.

Addition. The addition atom sums two expressions elementwise.

```
add_expr = cp.add(x, y)
```

The implementation below is the template every other atom follows. The forward pass evaluates the children and sums their values. `jacobian_init` allocates the output Jacobian, whose number of nonzeros is at most the sum of the children's, and fixes its pattern. `eval_jacobian` sums the children's Jacobians into it.

```
static void forward(expr *node, const double *u)
{   /* forward pass for children */
    node->left->forward(node->left, u);
    node->right->forward(node->right, u);
    for (int i = 0; i < node->size; i++)
    {
        node->value[i] = node->left->value[i] + node->right->value[i];
    }
}

static void jacobian_init_impl(expr *node)
{
    /* initialize children's jacobians */
    jacobian_init(node->left);
    jacobian_init(node->right);

    /* we never have to store more than the sum of children's nnz */
    int nnz_max = node->left->jacobian->nnz + node->right->jacobian->nnz;
    node->jacobian = new_csr_matrix(node->size, node->n_vars, nnz_max);

    /* fill sparsity pattern */
    sum_csr_alloc();
```

```
}

static void eval_jacobian(expr *node)
{
    /* evaluate children's jacobians */
    node->left->eval_jacobian(node->left);
    node->right->eval_jacobian(node->right);

    /* sum children's jacobians */
    sum_csr_fill_values();
}
```

Negation. The negation atom negates an expression elementwise. The forward pass flips the sign of each value. The Jacobian has the child's pattern, so `jacobian_init` copies it and `eval_jacobian` negates its values.

Transpose. The transpose atom transposes an expression.

```
transpose_expr = cp.transpose(x)
```

The forward pass writes the transpose into the row-major output buffer. Transposing only relabels the output entries, so the Jacobian is a fixed row permutation of the child's: `jacobian_init` records the permutation, and `eval_jacobian` copies the child's values into the permuted rows.

Horizontal stack. The horizontal stack atom concatenates expressions of equal row count side by side.

```
hstack_expr = cp.hstack([x, y])
```

The forward pass concatenates the arguments' values. The Jacobian stacks the arguments' Jacobian rows: `jacobian_init` lays out the combined pattern, and `eval_jacobian` copies each argument's values contiguously into place.

Vertical stack. The vertical stack atom concatenates expressions of equal column count.

```
vstack_expr = cp.vstack([x, y])
```

It needs no new code. The engine uses the identity `vstack([A, B]) = transpose(hstack([transpose(A) transpose(B)]))`, inheriting the forward and Jacobian behaviour of transpose and hstack.

Indexing. The indexing atom selects a subset of an expression's entries.

```
index_expr = x[0:3]
```

The forward pass gathers the selected entries. The Jacobian is the corresponding subset of the child's rows: `jacobian_init` records which rows are selected, and `eval_jacobian` copies their values.

Sum. The sum atom reduces all entries of an expression to a scalar.

```
sum_expr = cp.sum(x)
```

The forward pass accumulates the child's values into a scalar. The Jacobian is the column-wise sum of the child's Jacobian, a single row: `jacobian_init` fixes its pattern, and `eval_jacobian` accumulates the child's values into it.

Elementwise multiplication. The elementwise multiplication atom scales an expression entrywise by data `a`, constant or parameter.

```
elemwise_mult_expr = cp.multiply(x, a)
```

The forward pass multiplies the child's values by `a`, refreshing `a` first if it is a parameter. Scaling row i by a_i leaves the child's pattern unchanged, so `jacobian_init` copies it and `eval_jacobian` scales each entry of row i by a_i. A change in `a` is absorbed entirely at evaluation.

Matrix multiplication. The matrix multiplication atom takes a constant matrix $A \in \mathbf{R}^{m\times k}$ and an expression x with $x \in \mathbf{R}^k$, and produces $Ax \in \mathbf{R}^m$. The canonicalized problem is affine, so one operand is always data; we take it to be the left operand A, with the variable-bearing expression as the right child.

```
matmul_expr = A @ x
```

The forward pass computes the sparse matrix–vector product $y = Ax$. If x has Jacobian $J_x \in \mathbf{R}^{k\times n_{\text{vars}}}$, then y has Jacobian $J_y = AJ_x$, a sparse matrix–matrix product. Its pattern depends only on the patterns of A and J_x.

This is the one atom where the two-phase split does real work. We factor the product into a *symbolic* routine, which allocates J_y and fills its pattern from the patterns of A and J_x, and a *numeric* routine, which fills the values into that pattern. `jacobian_init` runs the symbolic routine once; `eval_jacobian` runs only the numeric routine on each solve. The symbolic

step is the expensive part of a sparse matrix–matrix product, and it is hoisted out of the per-solve path.

3.4 Problem class

The atoms compute the Jacobian of a single expression. A `Problem` ties together the full set of constraint expression trees and assembles their root Jacobians into the global problem data $[A\ b]$ and objective data $[c\ d]$.

At construction the problem runs the one-time *initialization* phase. It walks each expression tree once, calling `jacobian_init` bottom-up so every node allocates its Jacobian and fixes its pattern. The pattern of an affine expression is independent of its parameter values, so this walk runs once per problem family. The root Jacobians are concatenated into the global CSR buffer for A using the constraint row offsets of §2.2, fixing the pattern of the problem data as a whole.

3.5 Parameter updates

The *evaluation* phase runs on every solve. Given new parameter values θ, the problem refreshes the leaf parameters and calls `eval_jacobian` bottom-up, filling the values of each pre-allocated Jacobian without touching its pattern. The forward pass supplies the constant column b, and the root Jacobians are copied into the global buffer. A parameter update therefore costs only a forward pass and a numeric Jacobian evaluation: no allocation, no symbolic work, no graph mutation.

This is the saving over the `ignore_dpp` path of §2.2. That path is general — it places no DPP restriction on how parameters enter — but re-derives the symbolic structure on every change of θ. Our two-phase design keeps that generality while paying the symbolic cost once, recovering the re-solve speed of the cached DPP path without materializing the three-dimensional parameter tensor.

3.6 CVXPY interface

We expose the `Problem` class and the atom constructors through a thin set of Python bindings. The engine then acts as a *canonicalization backend*, occupying the place of `cvxcore` and the `SciPy` backend of §2.2: it consumes the canonicalized affine expression trees from the reductions and returns the problem data (A, b, c, d).

CVXPY hands the backend its canonicalized expression trees. The binding mirrors each

CVXPY atom onto its engine counterpart, builds the C expression trees, runs the initialization phase once, and stores the handle on the problem. On each solve with new parameter values, CVXPY calls the evaluation phase to obtain the updated data, which is forwarded to the cone solver. The binding forwards only pointers to pre-allocated buffers, so no problem data is copied between Python and C in the inner loop.

CHAPTER 4 Benchmark results

In this chapter we evaluate the engine of §3 against CVXPY's existing canonicalization path. The central question is the cost of repeatedly forming the problem data for a parameterized problem family, which is exactly the regime our two-phase design targets.

4.1 Benchmark Setup

We benchmark on the public CVXPY benchmark suite [80], 25 real-world problems (linear and quadratic programs, least squares, portfolio models, robust regression, and conic and semidefinite programs) maintained under *airspeed velocity* (`asv`) [81] with seeded, reproducible data. Three problems need the unsupported `kron` atom and are excluded; the remaining 22 run on both backends.

We time *canonicalization* only—the construction of the problem data (A, b, c, P) from the expression tree—and exclude the numerical solve, since that is what the engine changes. Concretely we time `get_problem_data` with the target solver fixed to Clarabel [82]: the `CPP` (`cvxcore`) baseline is the default DPP path, and the C engine is selected with `ignore_dpp`.

```
prob.get_problem_data(solver=cp.CLARABEL)              # CPP baseline (DPP
    tensor)
prob.get_problem_data(solver=cp.CLARABEL, ignore_dpp=True) # our C engine
```

We report two regimes: *cold-start* (a fresh `Problem` built with nothing cached, averaged over 3 builds) and *warm-restart* (parameter values reassigned on a warm cached chain, averaged over 20 re-compiles)—the latter being the regime the engine targets. Alongside these wall-clock times we report *peak memory*, and summarize across the suite with the *geometric mean* of the per-problem engine/`CPP` ratios, the standard scale-free statistic for timings that span orders of magnitude [83].

All measurements run single-threaded on a MacBook Pro (13-inch, M2, 2022): 8-core Apple M2, 24 GB RAM, macOS 26.3, with Python 3.14, CVXPY 1.10 (dev), NumPy 2.4 [30], and SciPy 1.16 [77]. Versions are pinned and the suite and engine are public, so the configuration is reproducible.

4.2 Results across the benchmark suite

Table 4.1 reports cold-start compilation across the 22 runnable benchmarks, and Table 4.2 reports warm-restart re-compilation on the parametric subset. In both, the ratio is C engine/CPP, so a value below 1 means the engine is faster; the final row gives the geometric mean of the per-problem ratios.

Table 4.1 Cold compilation time: a fresh problem formed and canonicalized from scratch, C engine versus CVXPY's `CPP` backend (22 runnable benchmarks). Ratio = C engine / `CPP`; values below 1 favor the engine.

Benchmark	`CPP` (ms)	C engine (ms)	Ratio
Yitzhaki	845	69	**0.08**
Cajas	2154	179	**0.08**
SimpleQP	2882	295	0.10
LeastSquares	2313	263	0.11
HuberRegression	2559	298	0.12
SimpleLP	5656	747	0.13
SlowPruning	1726	283	0.16
SVMWithL1Regularization	2593	411	0.16
TvInpainting	879	164	0.19
SimpleScalarParametrizedLP	1066	223	0.21
SimpleFullyParametrizedLP	463	110	0.24
ParametrizedQP	2645	631	0.24
ConeMatrixStuffing	2589	819	0.32
ParamSmallMatrixStuffing	118	40	0.34
OptimalAdvertising	2202	805	0.37
ParamConeMatrixStuffing	69	27	0.38
SmallMatrixStuffing	1633	627	0.38
CVaR	7128	6224	0.87
FactorCovarianceModel	1157	1117	0.97
Murray	1052	1627	1.55
ConvexPlasticity	81	220	2.63
SemidefiniteProgramming	995	8886	**8.90**
Geometric mean			**0.32**

Reading. In aggregate the engine is about 3× faster on cold builds (geometric-mean ratio 0.32) and near parity on parametric re-compiles (1.18), but the per-problem split is the more informative story. On *cold* compilation the engine wins on the bulk of the LP, QP, portfolio, and conic problems—typically 3–12× faster, since it extracts the problem data directly from

Table 4.2 Parametric re-compilation time on a warm cached chain: parameter values are reassigned and the problem data rebuilt (the engine's target regime). Ratio = C engine / `CPP`.

Benchmark	#par	`CPP` (ms)	C engine (ms)	Ratio
ParamSmallMatrixStuffing	3	2.23	0.65	**0.29**
ParamConeMatrixStuffing	3	1.19	0.40	0.33
SimpleFullyParametrizedLP	1	50.8	36.8	0.72
ParametrizedQP	2	333	309	0.93
SimpleScalarParametrizedLP	1	66.2	67.9	1.03
FactorCovarianceModel	2	21.3	72.2	3.39
SVMWithL1Regularization	1	40.2	150	3.72
ConvexPlasticity	3	43.0	194	4.52
Geometric mean				**1.18**

the expression trees while `CPP` pays the full parameter-tensor build. It loses on `Murray` (1.55×), `ConvexPlasticity` (2.63×), and the `SemidefiniteProgramming` outlier (8.9×), and reaches parity on `FactorCovarianceModel` (0.97×). On *parametric re-compilation* the picture is mixed and use-case dependent: the engine is faster on the matrix-stuffing and fully-parametrized-LP re-solves (0.29–0.72×) and near parity on `ParametrizedQP`, but 3.4–4.5× slower on `FactorCovarianceModel`, `SVM`, and `ConvexPlasticity`, where `CPP`'s cached DPP parameter→data tensor (one sparse matmul per solve) is hard to beat. The largest memory wins come from keeping constants sparse: eliminating an eager $25\,000 \times 25\,000$ dense densification (≈ 5 GB) on `FactorCovarianceModel`, and shrinking the `ConvexPlasticity` Hessian block from 9M to 51k nonzeros by routing a mostly-zero `quad_form` matrix to the sparse binding.

4.3 Scaling with problem size

The cross-suite tables fix the problem size; here we instead sweep it. We deep-dive three parameterized problems—parametric least squares, a factor-covariance portfolio, and an energy-storage scheduling model—and re-measure cold-start time, warm-restart time, and peak memory as the size grows. All three are genuinely parametric, so the warm-restart curve measures the engine's target regime: refilling values on a fixed sparsity pattern as the parameters change. For each panel we plot `CPP` against the C engine on log–log axes.

4.3.1 Parametric least squares

The bread-and-butter case: fit x to noisy observations with the data matrix A and right-hand side b supplied as parameters, so a re-solve only reassigns A and b. The size knobs are the number of observations m and variables n.

```
import cvxpy as cp

A = cp.Parameter((m, n))    # data matrix  (reassigned each solve)
b = cp.Parameter(m)         # observations (reassigned each solve)

x = cp.Variable(n)
objective  = cp.Minimize(cp.sum_squares(A @ x - b))
constraints = [0 <= x, x <= 1]
prob = cp.Problem(objective, constraints)
```

We sweep $n \in \{250, 500, 1000, 2000, 4000\}$ with $m = 3n$ (overdetermined) and measure both backends in the *ignore-DPP* regime: the canonicalization is rebuilt from scratch whenever the parameters change, which is the engine's intended use case. The C engine carries the parameters natively and re-extracts values on a fixed sparsity pattern; the `CPP` baseline substitutes the parameter values (the `EvalParams` reduction) and re-canonicalizes the resulting constant-data problem. We deliberately avoid CPP's DPP parameter-tensor path here: because the data matrix A is itself an $m \times n$ parameter, that tensor has $\Theta(mn)$ entries and its construction dominates everything else—tens of seconds and gigabytes already at $n = 500$—so it is neither representative nor the comparison of interest. Cold-start and warm-restart times are medians over three repetitions; peak memory is the process resident-set high-water mark (`ru_maxrss`), measured in a fresh subprocess per point so that, unlike Python-level `tracemalloc`, it also captures the engine's C-side allocations.

Cold-start (Figure 4.1) favours the engine by roughly 4× across the range (ratio 0.21–0.27), both backends scaling as $\sim n^{2.1}$. The gap widens on warm-restart (Figure 4.2): `CPP` re-canonicalizes in essentially its cold-start time (9.1 s at $n = 4000$), whereas the engine re-extracts on the cached pattern in 863 ms—a 10–12× speed-up that is precisely the non-DPP re-solve regime the engine targets. Peak memory (Figure 4.3) is comparable: both grow as $\sim n^{1.6}$ and are dominated by the dense A, with the engine 2–30% heavier in resident set. Its own internal accounting (the engine's `verbose` "peak memory") grows as $\sim n^{2.0}$, from 14 MB at $n = 250$ to 3.7 GB at $n = 4000$, yet stays below the process RSS throughout.

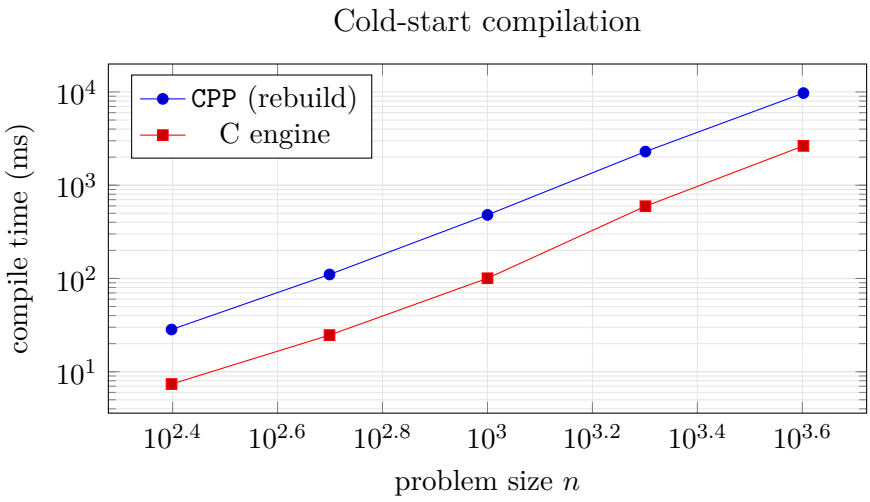


Figure 4.1 Parametric least squares, cold-start: full canonicalization from scratch versus problem size n ($m = 3n$). The C engine is $\approx 4\times$ faster across the whole range; both backends scale as $\sim n^{2.1}$.

4.3.2 Factor-covariance portfolio

A Markowitz portfolio whose risk is expressed through a low-rank factor model: n assets load on $m \ll n$ factors with covariance Σ, plus idiosyncratic diagonal risk D. The risk-aversion γ and leverage limit $L_{\max}$ are parameters, so re-solving traces the efficient frontier without rebuilding the problem. The size knobs are the number of assets n and factors m.

```
import cvxpy as cp
import numpy as np

# mu: expected returns, F: factor loadings (n x m),
# Sigma: factor covariance (m x m), D: idiosyncratic variances (diagonal).
w = cp.Variable(n)                # portfolio weights
f = cp.Variable(m)                # factor exposures
gamma = cp.Parameter(nonneg=True) # risk aversion (reassigned each solve)
Lmax  = cp.Parameter()            # leverage limit (reassigned each solve)

ret  = mu.T @ w
risk = cp.quad_form(f, Sigma, assume_PSD=True) + cp.sum_squares(np.sqrt(D) @ w
    )
objective  = cp.Maximize(ret - gamma * risk)
constraints = [cp.sum(w) == 1, f == F.T @ w, cp.norm(w, 1) <= Lmax]
prob = cp.Problem(objective, constraints)
```

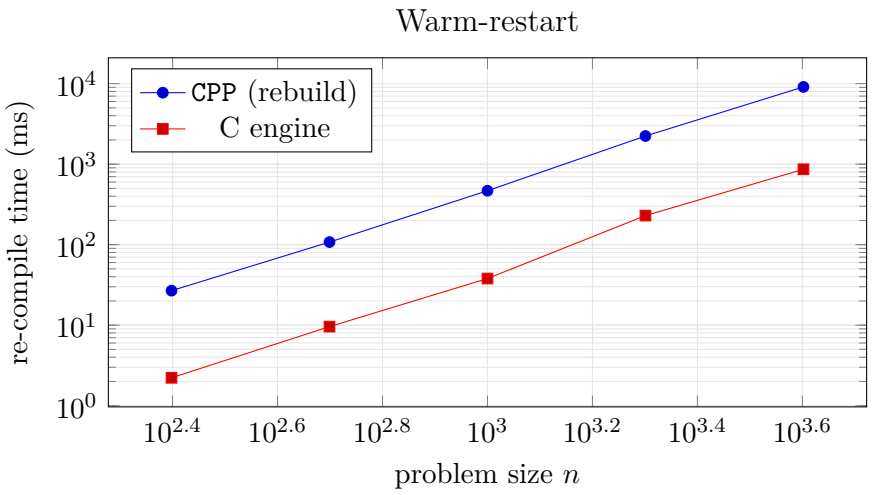


Figure 4.2 Parametric least squares, warm-restart: rebuild after reassigning A and b. With no DPP tensor to reuse, `CPP` pays its full cold-start cost again, while the C engine only re-evaluates values on the cached sparsity pattern—a 10–12× speed-up.

We hold the factor count fixed at $m = 100$ and sweep the assets $n \in \{500, 1000, 2000, 4000, 8000\}$. As in the least-squares study, both backends run in the *ignore-DPP* rebuild regime—the engine carries $\gamma, L_{\max}$ natively and re-extracts on a fixed sparsity pattern, while `CPP` substitutes their values and re-canonicalizes. (Here the parameters are scalar, so CPP's DPP cache would in fact be feasible; we use the same rebuild regime as before for a consistent comparison.) Times are medians over three repetitions; peak memory is the process resident-set high-water mark measured in a fresh subprocess.

Cold-start (Figure 4.4) again favours the engine, but less decisively than for least squares and with a narrowing margin: from 0.41× at $n = 500$ to 0.88× at $n = 8000$, as the engine's $\sim n^{1.2}$ cold cost catches CPP's $\sim n^{0.9}$. Warm-restart (Figure 4.5) is where the engine dominates: `CPP` pays its full cold-start cost on every re-solve (144 ms at $n = 8000$), while the engine refills the cached pattern in 10.7 ms—a steady 12–13× advantage. Peak memory (Figure 4.6) is close, the engine 4–16% heavier in resident set; both curves are nearly flat because the low-rank model keeps the operator small. The engine's own internal peak (its `verbose` accounting) grows as $\sim n^{0.9}$, from 6 MB at $n = 500$ to 74 MB at $n = 8000$.

4.3.3 Energy-storage scheduling

A simple energy problem: schedule a battery over a horizon of T hours to minimize the cost of grid energy, where the hourly price is the parameter that changes between solves. The state of charge is the running sum of charge minus discharge, and the battery must end

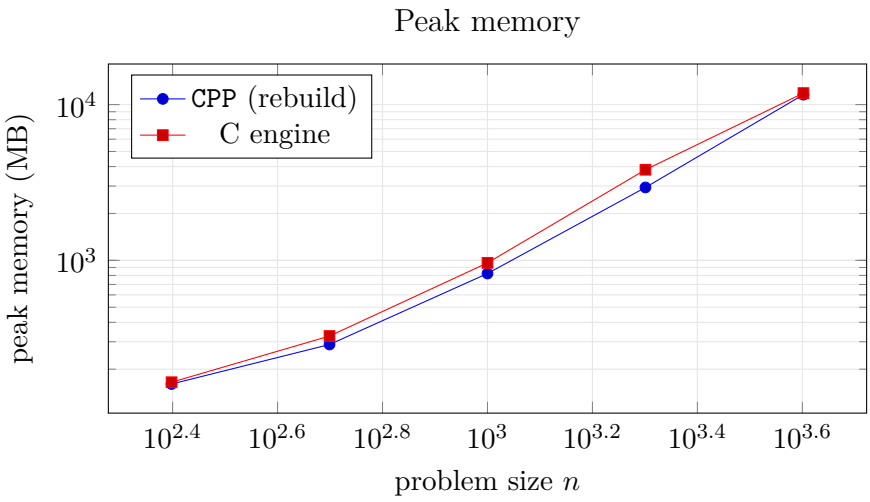


Figure 4.3 Parametric least squares, peak process memory (resident-set high-water mark). The backends are within ∼30% of each other; both grow as $\sim n^{1.6}$, dominated by the shared interpreter and the dense data matrix A.

empty. The size knob is the horizon T.

```
import cvxpy as cp
import numpy as np

price = cp.Parameter(T)          # hourly price (reassigned each solve)
load  = np.random.rand(T)        # fixed demand profile
Q, Pmax = 50.0, 20.0             # capacity and charge/discharge power limit

charge    = cp.Variable(T, nonneg=True)
discharge = cp.Variable(T, nonneg=True)
soc = cp.cumsum(charge - discharge)     # battery state of charge

constraints = [charge <= Pmax, discharge <= Pmax,
               soc >= 0, soc <= Q, soc[T - 1] == 0]
grid = load + charge - discharge        # net energy drawn from the grid
prob = cp.Problem(cp.Minimize(price @ grid), constraints)
```

The cross-suite results (§4.2) and these scaling curves together delineate the operating regime where the engine pays off: cheap from-scratch builds across the suite, and a favorable size trend on problems whose canonicalization cost the engine avoids re-paying.

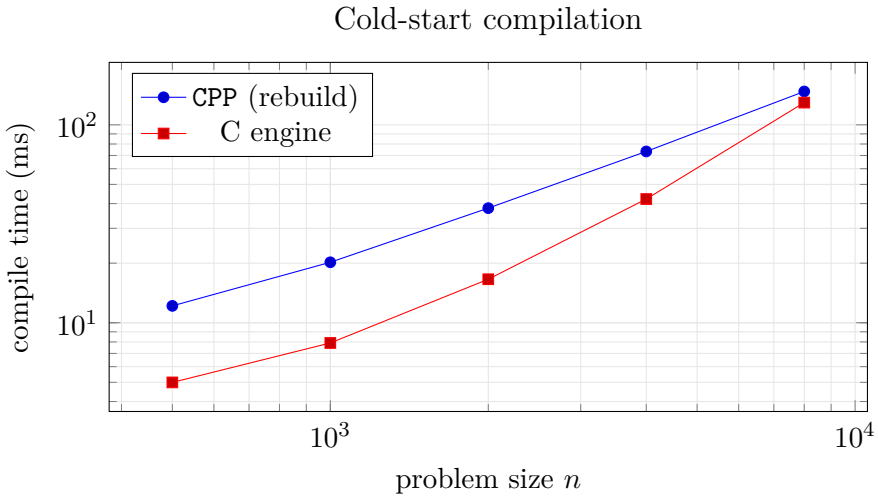


Figure 4.4 Factor-covariance portfolio, cold-start versus number of assets n ($m = 100$ factors). The engine leads at small n but its advantage shrinks with size—it scales as $\sim n^{1.2}$ against CPP's $\sim n^{0.9}$, reaching near-parity by $n = 8000$.

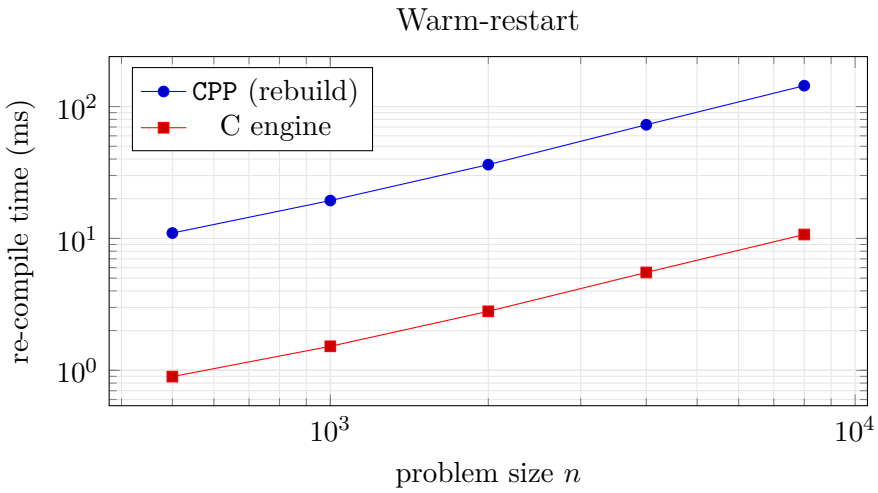


Figure 4.5 Factor-covariance portfolio, warm-restart after reassigning γ and $L_{\max}$. `CPP` re-canonicalizes in essentially its cold-start time; the engine re-extracts on the cached pattern, a stable 12–13× speed-up across the range.

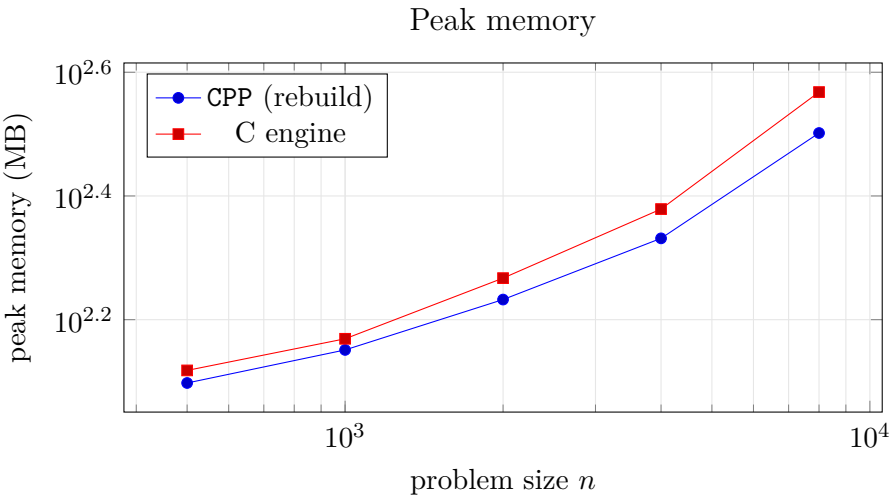


Figure 4.6 Factor-covariance portfolio, peak process memory. The backends are within $\sim$15%; both grow slowly ($\sim n^{0.35}$), reflecting the low-rank factor structure rather than a dense operator.

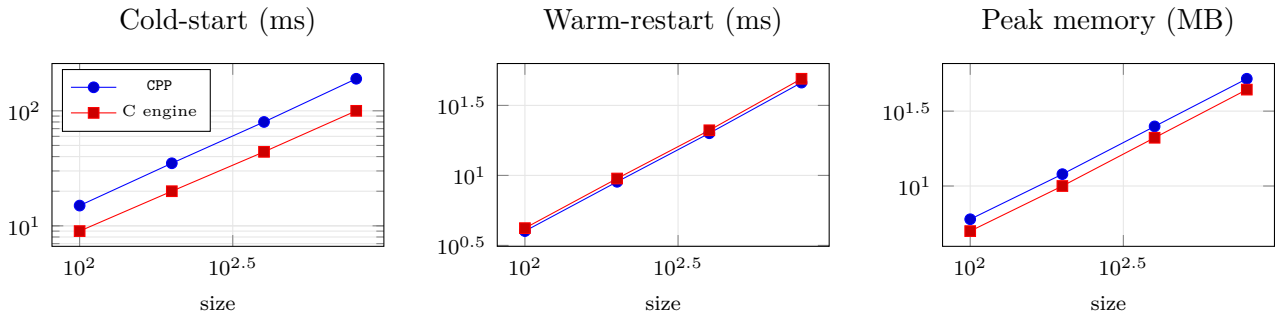


Figure 4.7 Energy-storage scheduling: scaling of cold-start, warm-restart, and peak memory with horizon length T (*placeholder curves; data pending the size-sweep run*).

CHAPTER 5 CONCLUSION

5.1 Summary of Works

5.2 Limitations

5.3 Future Research

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